Challenge 3

Sustainable Sonar Systems for Marine & Climate Protection

Detailed Participant Guide

EGYPT SECTION CHAPTER

JORDAN SECTION CHAPTER

TUNISIA SECTION CHAPTER

ESSTHS IEEE Student Branch

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Detailed Participant Guide

Challenge Overview

- Design a sonar-based sensing concept for marine or climate-related applications such as coral reef monitoring, marine debris detection, pollution tracking, or seabed observation.
- Your solution must maintain useful sensing capability while reducing acoustic disturbance, limiting unnecessary operation, and improving environmental sustainability.
- Physical sonar hardware is not required. A strong concept, clear system logic, and credible signal-processing simulation are sufficient for evaluation.

1. Scope and Topic

This challenge addresses the need for marine sensing systems that are both technically useful and environmentally responsible. Conventional sonar approaches can provide valuable information for monitoring reefs, debris, pollution, or habitat changes, but poorly designed acoustic operation can increase disturbance to marine ecosystems. Participants are expected to propose a sonar concept that improves environmental monitoring while explicitly reducing acoustic impact and unnecessary energy use.

Participants may build their solution around one clearly defined application, such as:

- Coral reef or shallow-water habitat monitoring
- Marine debris or object detection in coastal waters
- Pollution plume observation or environmental survey
- Low-impact seabed or habitat observation

2. Problem Statement

IMarine sensing is often needed in environments where direct visual monitoring is limited, water conditions are variable, and wide-area coverage is difficult. Sonar remains one of the most practical sensing methods in such scenarios, but aggressive acoustic transmission can produce unnecessary disturbance, especially when systems operate continuously, transmit at higher-than-needed source levels, or use poorly justified operational settings. A sustainable sonar system must therefore answer two questions at the same time:

- How can the system still sense the target or environment effectively?
- How can the system reduce harm, disturbance, and wasted operation compared with a conventional approach?

3. Challenge Goal

Create a validated sonar concept that supports a real marine or climate-related sensing task while demonstrating an eco-friendly operational strategy. The proposed system should combine a suitable sonar architecture, a clear signal-processing chain, and a defensible sustainability strategy. Your work should show how sensing performance is balanced against reduced acoustic impact, lower or more selective transmission, and more responsible deployment logic

Your submission should make all three of the following visible:

- What the sonar is sensing and where it operates.
- How the system works technically, including transmission, reception, and processing logic.
- How the system reduces environmental impact compared with a more conventional sonar configuration.

4. What Participants Are Expected to Build

- A sonar system concept focused on one clear use case.
- A signal-processing and simulation model using MATLAB or Python.
- An environmental operating strategy that reduces acoustic impact and explains trade-offs.
- Evidence through analysis, plots, comparisons, or simulation outputs.

You are not required to build physical sonar hardware. A well-documented simulation, supported by sound assumptions and meaningful comparisons, is fully acceptable.

5. Technical Instructions

5.1 Sonar Type & Operational Mode

Define the sonar architecture clearly. You may choose active, passive, or hybrid operation, but you must justify the choice in relation to the marine scenario and the sustainability objective.

- Selected sonar type: active, passive, or hybrid.
- Conceptual operating frequency range.
- Transmission mode: pulse, continuous, intermittent, adaptive, or listening-based.
- Operational duty cycle and conditions for activation.

5.2 Marine Scenario Definition

Describe the sensing environment in practical terms so the jury can understand what the system is trying to detect, under what conditions, and why sonar is suitable.

- Marine environment: reef, coastal, open ocean, shallow water, harbor, or survey zone.
- Target or phenomenon: coral structures, debris, pollutants, seabed features, habitat indicators, or environmental change.
- Deployment model: fixed platform, mobile platform, short survey mission, or periodic observation.
- Expected operating frequency: continuous, event-based, scheduled, or adaptive.

5. Technical Instructions (Cont.)

5.3 Sustainability Strategy

This is the core of the challenge. Sustainability must be deliberate, explicit, and technically justified. Avoid vague statements such as “the system is green” unless you show exactly why.

- Reduced source level compared with a conventional case.
- Limited duty cycle or shorter transmission windows.
- Passive listening or hybrid operation whenever possible.
- Adaptive transmission settings based on environment or task needs.
- Lower-power operation or more selective use of sensing resources.

5.4 Justify component selection

Your simulation must show that the concept is technically meaningful. It should represent the main sensing logic and allow the jury to see how your eco-friendly choices influence performance.

- Signal generation and reception model.
- Noise and propagation assumptions suitable for the chosen environment.
- Detection, classification, imaging, or estimation logic as relevant.
- Comparison between a conventional configuration and your eco-friendly configuration.
- Trade-off analysis between sensing performance and environmental protection.

5.5 Validation Expectations

- Use plots, detection outputs, comparison charts, or tables of key assumptions and results.
- Explain why your chosen parameters are credible, even if they are conceptual.
- Show what is gained and what is sacrificed when reducing acoustic impact.
- Keep the model focused and understandable rather than overly complex.

6. Online Phase Deliverables

6.1 System Concept Document (PDF)

This is the main written explanation of your project. It should be concise, technically structured, and easy to review.

- Problem statement and marine monitoring use case.
- Sonar type, operational mode, and conceptual architecture.
- Marine scenario and deployment context.
- Sustainability strategy and comparison logic.
- High-level processing flow and expected outputs.

6.2 Signal Processing / Simulation Files

- MATLAB or Python source files.
- Simulation notebooks or scripts.
- Plots, generated results, and comparison outputs.
- Any supplementary notes needed to reproduce the main results.

6.3 Demo Video (2–3 Minutes)

The demo video should prove that the project is coherent, technically grounded, and environmentally justified.

- Introduce the marine problem and why sonar is relevant.
- Explain the sonar concept and key operating choices.
- Show the simulation setup and outputs.
- Highlight how acoustic impact or unnecessary operation is reduced.
- Present at least one clear comparison or trade-off.

6.4 Final Presentation Deck (PDF)

- Marine problem overview.
- System concept and architecture.
- Signal-processing approach.
- Key results and comparison outputs.
- Sustainability justification and final takeaways.

7. Repository & Submission Standards

All teams should maintain a clean project repository and submit files using the official Phase 1 structure. The repository strengthens technical transparency and helps judges verify implementation quality.

Repository Section	What It Should Contain
README.md	Project overview, challenge title, team name, setup instructions, sustainability explanation, and how to run the main simulation.
/docs	Architecture diagrams, concept notes, marine scenario definitions, parameter tables, and supporting explanations.
/src/ or /simulation/	MATLAB scripts, Python notebooks, signal models, detection logic, and processing pipeline files.
/results/	Plots, comparison charts, output figures, and key quantitative results.
/assets/ (optional)	Illustrations, diagrams, demo visuals, or media assets used in the presentation.

- File naming convention should follow the Phase 1 structure: TeamName_ChallengeTitle_Phase1_FileType.
- Repository access should be public or jury-accessible.
- If AI tools are used for code, optimization, or documentation support, that usage should be stated clearly and all outputs must be validated and explainable.
- Judges should be able to reproduce the key results from the provided files and instructions.

8. Tools You May Use

Simulation & Coding Tools

- MATLAB
- Python (NumPy, SciPy, Matplotlib)
- Jupyter Notebook
- Simulink (optional)

AI Tools Policy

- AI tools may be used for code assistance, visualization support, or idea refinement.
- You must understand and validate all AI-assisted outputs.
- You must be able to explain your technical choices during judging and finals.

9. Finals (Onsite Phase)

If selected as a finalist, your team will present the project in a live session and defend both the technical design and the environmental logic of the sonar concept.

- 7-minute pitch covering the problem, concept, simulation, and sustainability strategy.
- Live or recorded interactive visualization of the signal behavior, outputs, or comparison cases.
- Focused judge discussion on acoustic safety, realism of assumptions, and trade-offs.

Judges may ask questions such as:

- How does your system reduce disturbance to marine life?
- Why are your chosen frequencies or source levels safer or more appropriate?
- What performance trade-offs did you accept in order to improve sustainability?
- How realistic are your assumptions about propagation, targets, and environment?

10. Judging Rubric

Phase 1 - Online Evaluation

Criterion	Weight	What judges look for
Concept Realism	25%	Feasible sonar architecture and credible operating concept.
Marine Sustainability Strategy	30%	Clear, explicit reduction of acoustic impact and environmental disturbance.
Signal Processing Quality	20%	Sound modeling, processing logic, and technical implementation.
Simulation Evidence	15%	Clear outputs, comparisons, and supporting evidence.
Documentation Quality	10%	Professional structure, clarity, and completeness of submitted materials.

Phase 2 - Finals Evaluation

Criterion	Weight	What judges look for
Sustainability Defense	30%	Ability to justify marine ecosystem protection claims.
Technical Feasibility Discussion	25%	Realistic acoustic assumptions and sound engineering defense.
Live Simulation Demonstration	20%	Interactive clarity and credibility of outputs.
Environmental Trade-off Awareness	15%	Understanding of performance versus safety trade-offs.
Presentation Quality	10%	Structured, clear, and confident communication.

11. Tips for Strong Submissions

- Focus on one marine use case and explain it deeply.
- Show a clear comparison between a conventional configuration and your eco-friendly approach.
- Use realistic assumptions rather than highly ambitious but unsupported claims.
- Explain the trade-off between sensing effectiveness and environmental protection.
- Keep the signal-processing chain understandable and well documented.

12. Common Mistakes to Avoid

- Ignoring marine ecosystem impact and treating sustainability as an afterthought.
- Using aggressive sonar parameters without justification.
- Providing no comparison between standard and eco-friendly operation.
- Submitting unclear plots or poorly explained signal-processing logic.
- Making broad environmental claims without technical support.

13. Final Advice to Participants

A strong submission answers this question clearly: How can we sense the ocean while respecting and protecting marine life? If your concept shows a realistic sensing task, a sound processing approach, and a defensible reduction in acoustic impact, then you are well aligned with Challenge 3.

Final Submission Checklist

• GitHub repository or equivalent project folder with code, documentation, and outputs.
• System Concept Document (PDF).
• Signal Processing / Simulation files.
• Demo Video (2–3 minutes).
• Presentation Deck (PDF).
• Consistent naming and clear instructions for reviewers.